

Supplementary Information

Why can equal results lead to different scientific futures

How humans and AI jointly form the conditions for further discovery

Supplementary Note 1. Claim architecture

The manuscript combines a theoretical target, an analytical witness and a local recorded application. They do not supply three replications of one mechanism. The title concerns equivalence under a declared present observation and non-equivalence under a declared future regime. It does not assert that identical complete causal states yield different outcomes. “Scientific futures” includes testing, correction, question formation, method development and adaptation under specified duties, tools, permissions and resources; these components need different observations. The proof-completion pair observes local application and supplied downstream use. Separate archived coefficient studies observe actual changed-target revision and reuse; their aggregate policy effects remain unestablished.

The theory takes problem–method–evidence–division-of-work relations as an explanatory object. Some relations are instantiated by a proof interface or a parameter map; others concern responsibility and access. They can be included in a classical state model. The proposal is not an irreducible force outside such a model. Its target is the formation and reconstruction of conditions by activity, rather than assuming those conditions to be permanently supplied from outside.

A minimal evidence discipline for a relation has four parts: identify its carrier; specify the operation for which it matters; observe its actual use or specify a proof of dependence; and distinguish how it was formed from how it was handed over. Adding an omitted variable to a diagram does not establish these parts. Conversely, an intervention need not isolate every internal psychological mechanism to establish a bounded total effect. The proof-completion pair and post-selected successor traces establish particular objects, changes and uses. They do not establish a stable average benefit: the complete coefficient-study results include identical versions, empty terminal outputs and execution-dependent contrasts.

Supplementary Table 1 | Evidence roles and claim boundaries.

Evidence object	Established content	Its precise contribution to the mother question	What it does not establish
Coefficient witness	A specified readout cannot support exact correction at two admissible inherited states; a positive minimax lower bound	Shows why a retained consequence of prior action can be relevant beyond the current summary	How a human or model acquired that state; equal complete current function
Safe state-aware construction	Compact information can remove that local obstruction under explicit domain assumptions	Demonstrates a sufficient alternative to retaining all history	Global PDE admissibility of arbitrary code; historical indispensability
Recorded coefficient continuation	Actual programmes revised after feedback, inherited fields received by later work, and renewed numerical action	Observes a restricted form of reconstruction and reuse	A stable average assistance/adoption benefit, a unique internal mechanism, or the full NS proof
hclose application	A particular lemma application is constructed, accepted in retained compilation and used by a fixed consumer	Makes one conversion from available result to operative condition inspectable	Autonomous formation of the consumer, full U3 or the complete practice cycle

Evidence object	Established content	Its precise contribution to the mother question	What it does not establish
Human task revision	The investigator changed scope, materials and procedure during exploration	Records human organisation as part of the process	An independent measure of the investigator's understanding or its improvement
Domain predictions	Prospective conditional tests of result/continuation separation	States how wider implications can be refuted or narrowed	Existing evidence for all human, institutional or disciplinary claims

Supplementary Note 2. Exact finite separation

2.1 Original comparison and analytical scope

This argument is inherited from the preserved R6 mathematical note of 18 September 2026, with its assumptions retained. The original note remains in the authors' internal historical provenance; the complete argument and its assumptions are reproduced here and mapped to the executable checks in REPRODUCIBILITY/math/README.md. In a local covariance reduction, the actual amplitude relative to a fixed primary coefficient is z , the target squared amplitude is p , and the current reported residual is $r = p - z^2$. The update is a real increment g and the complete residual is $E = (z + g)^2 - p$. The stored or inherited state is not reset between operations. With $u = z - 1$,

$$E = 2g + g^2 - r + 2ug.$$

This is an exact identity, not a behavioural model of the language model. It separates the residual of a correction at the fixed base from its cross term with the actual state already produced. The local reduction inherits a role from the source covariance construction, but it removes other components of the full PDE problem. A small value here does not prove a small velocity, pressure or full Navier–Stokes residual.

2.2 Impossibility for a source-only action

Assume $0 < r \leq 1/4$ and $0 < v \leq 1/8$. Define

$$z_+ = 1 + v, z_- = 1 - v, p_{\pm} = z_{\pm}^2 + r.$$

An action observing only the same r chooses the same scalar g . It then has residuals

$$E_{\pm} = g^2 + 2g - r \pm 2vg.$$

If $E_+ = E_- = 0$, subtraction gives $4vg = 0$. As $v > 0$, $g = 0$, which gives $E_{\pm} = -r \neq 0$, a contradiction. What is equal is the declared summary r , not the targets $p_{\pm}$, the actual states or the full causal conditions. The choice of a deterministic source-only action is a stated restriction, not a description inferred from model outputs.

2.3 Exact minimax value

For any real numbers A, B , $\max(|A + B|, |A - B|) = |A| + |B|$. Hence

$$F(g) := \max(|E_+|, |E_-|) = |g^2 + 2g - r| + 2v|g|.$$

Put $g_0 = \sqrt{1+r} - 1 > 0$. For $g \geq 0$, the quadratic term changes sign only at g_0 . For $0 < g < g_0$,

$$F'(g) = -2g - 2 + 2v < 0,$$

and for $g > g_0$,

$$F'(g) = 2g + 2 + 2v > 0.$$

Thus F is minimised on that half-line at g_0 , with value $2v g_0$. For $g \in [-2, 0]$, $g^2 + 2g \leq 0$, so $F(g) \geq r$. Since $r = g_0(g_0 + 2) > 2v g_0$, this interval cannot improve the minimum. For $g \leq -2$, write $h = -g - 2 \geq 0$. Then $g^2 + 2g = h^2 + 2h$ and

$$F(g) = |h^2 + 2h - r| + 2vh + 4v = F(h) + 4v.$$

Consequently

$$\inf_{g \in \mathbb{R}} F(g) = 2v(\sqrt{1+r} - 1).$$

This piecewise proof establishes the whole stated range. The accompanying deterministic curve illustrates the analytical result; it is not its proof. This is elementary analysis applied to a specific continuation problem, not a claim of priority for minimax reasoning.

2.4 Why the one-variable repair can miss a leading term

Consider the earlier restricted interface

$$g = r/2 + \chi(r)r^2 B(r),$$

where B is fixed and smooth near zero and $\chi = 1$ there. If $u = a\tau + O(\tau^2)$ and $r = b\tau^2 + O(\tau^3)$, then $r^2 B = O(\tau^4)$, while the leading complete residual is

$$E = ab\tau^3 + O(\tau^4).$$

The state-dependent cross term is therefore not generally removable by that fixed interface. This does not exclude τ -dependent families, finite interpolation, other algorithms or richer state representations. It does not retroactively classify an earlier finite model answer as incorrect.

2.5 A compact sufficient repair

Let χ_r and χ_u be fixed smooth cutoffs, with inner radii $1/4$ and $1/8$, and outer radii $1/2$ and $1/4$, respectively. Let $\Phi = \chi_r \chi_u$ and

$$g = r/2 + \Phi[r^2 B(r, u) + ur C(r, u)].$$

On the closed outer rectangle, $z = 1 + u \geq 3/4$ and $z^2 + r \geq 1/16$. Set $w = \sqrt{z^2 + r}$ and

$$C_* = -\frac{1}{2z}, B_* = -\frac{1}{2z(w+z)^2}.$$

These functions extend analytically to a neighbourhood of the rectangle. Rationalisation gives

$$w - z = \frac{r}{w+z} = \frac{r}{2z} - \frac{r^2}{2z(w+z)^2},$$

and $r/(2z) = r/2 - ur/(2z)$. Thus where $\Phi = 1$, $g = w - z$ and $E = 0$. This formula supplies an actual sufficient alternative to a source-only action. The compact pair (r, u) and the stated rule suffice for the local operation; an entire exploratory transcript is not mathematically necessary.

For general fixed smooth B, C in this family, let $h = r^2 B + ur C$. Then

$$E = ur + r^2/4 + \Phi(2 + 2u + r)h + \Phi^2 h^2.$$

Under the scaling above,

$$E = ab[1 + 2C(0, 0)]\tau^3 + O(\tau^4).$$

This is a conditional prediction about an explicit rule, not a claim that the experimental model learned $C(0, 0) = -1/2$ or that the later hclose task tests this coefficient.

2.6 Boundaries that must remain

The state-aware core formula does not assert an improvement in every norm outside that core. The earlier counterexample $z = 9/10$, $r = 1/5$ gives complete residual $-1/100$ for $g = r/2$ and $1/5 - \sqrt{30}/25$ for $g = \sqrt{1+r} - 1$. The latter has zero pure residual but a larger complete absolute residual. This is an analytical counterexample already present in the supplied history, not a newly run experiment.

The source weighted extension additionally assumes fixed-stage derivative bounds, smooth cutoffs, permissible slow-variable multipliers and flat edge weights. The constants may depend on the actual chosen functions. Neither finite evaluation nor a local analytic-domain certificate validates every full-PDE condition. Earlier source-relative arguments for full-cycle compatibility are preserved as those arguments, not upgraded to external independent review in this manuscript.

Likewise, changing a legitimate representation need not change its physical action. Under a compatible positive slow rescaling of the wave basis,

$$b'_i = \lambda_i b_i, H' = H \text{diag}(\lambda^2), y' = y / \lambda^2, d' = d / \lambda^2,$$

the relative source $r = d / y$ is unchanged. With the corresponding amplitude transformation, the represented action is unchanged. The required multiplier and inverse-derivative bounds are part of the statement. Arbitrary fast or unbounded rescaling is not licensed by algebra alone.

Supplementary Note 3. Sufficient continuation

3.1 A declared future regime

Write F for a declared set of future duties, tools, permissions, resource rules and admissible continuation policies. Let $M(x)$ be the present output under comparison and $C(x)$ a candidate compact continuation state. Product equality $M(x) = M(x')$ does not by definition impose equality of C or of future trajectories. Conversely, complete causal-state equality with identical transition and randomness laws gives the same future law. “History” explains a present difference only through a carrier or a boundary condition included in the model.

The theoretical use of “scientific future” here is deliberately conditional. It can include a new verification duty, a changed assumption, an opportunity to formulate a problem, or a method-reconstruction task. Equality on one finite set of tests does not prove equality on all future science. In particular,

psychological understanding and the quality of generated scientific questions require their own operationalisations; they are not read out by the current compiler.

3.2 Conditional sufficiency proposition

Suppose that, for each permitted action and future input in F , (i) the law of the relevant observation and its cost depends on x only through $C(x)$; (ii) the next compact state can be updated from the present compact state, action and observation, with the same transition law; and (iii) the compared policy chooses the same distribution of actions when supplied the same compact state and observed history. Then systems with equal initial compact states have the same joint distribution of observations, costs and compact successor states for every finite horizon covered by F .

Proof. At the first step, condition (iii) gives equal action laws; condition (i) then gives equal observation and cost laws. Condition (ii) gives equal successor-state laws. Applying the same three statements conditional on the resulting shared history gives the next-step equality. Induction establishes the finite-horizon result. The statement is a standard sufficient-state composition principle, not a new result in probability or control theory.

The assumptions include the updates, not merely the prediction of the next answer. If duty changes reveal information not represented by C , the assumptions may cease to hold. If a decoder or a rederivation procedure reconstructs the required state but uses extra time or queries, outcome sufficiency may hold while budget-constrained equivalence does not. A claim of cost equivalence therefore includes construction, retrieval, compaction and restoration costs. Unobserved failure or missing events cannot be inserted as zero-cost steps.

3.3 Relation to a proof interface

If an existing result has type $L : \forall \theta, A(\theta) \rightarrow C(\theta)$ and a continuation consumes I , it is sufficient to preserve or reconstruct a valid parameter choice, a proof of its assumptions and a valid map from the resulting conclusion to I . Preserving the whole historical conversation is one possible carrier, not a logical requirement. The policy may also find another legitimate route to I . This statement describes an ordinary interface of proof composition; it does not explain by itself how that interface was learned, selected or socially allocated.

The mother question becomes empirical when those application relations, state records, checking responsibilities or reconstruction procedures are themselves outcomes of preceding activity. Their causal attribution requires real formation and use records, plus an appropriate comparison. The analytical witness specifies one dependency; actual coefficient programmes instantiate its whole-residual algebra in continuation records. The `hclose` case concerns a different application interface. No equality between their latent mechanisms is asserted.

Supplementary Note 4. The `hclose` mathematics

4.1 Identity and notation

The experimental source is a retained copy of `NavierStokes/PrimaryCovarianceBounds.lean` from the stated OpenAI commit. The larger source result concerns a forced three-dimensional incompressible Navier–Stokes construction. This manuscript does not independently validate its complete global theorem, nor claim the unforced regularity problem has been re-proved here. The empirical endpoint is the enclosing local type `checkpoint_bridge_goal` after insertion of the `hclose` proof body.

Supplementary Table 2 | Notation and source identifiers.

Manuscript notation	Original identifier or meaning
R_n	slotRadius r0 h n
L_n	ChartScales.slotLength r0 h n
$\overline{H}(P)$	normalizedPair P
$H_0(\xi), T_0(\xi)$	Reference matrix and target; original parameter name p
e, d	Source coefficients E,D, both non-negative
c	PulseCovariance.concentrationConstant a b B
Q	$E + D * \text{abs } c$
ρ, δ, M	Uniform tolerance, positive margin and entry bound from the supplied compactness result
S_n	Real.sqrt (ChartScales.S n)
c_j	Positive column-mass factors, distinct from C

4.2 Complete transport of the premise

The available local hypothesis is

$$\left| \frac{P_j^t(v)_i}{P_j^x(v)} - H_0(\xi)_{ij} \right| \leq \frac{e}{L_n} + d \frac{|v - L_n/2|}{L_n}, v \in [0, L_n].$$

The integral estimate expects the same assertion with R_n^2 everywhere in place of L_n . The provided `slotRadius_sq` establishes their equality. For a given $v \in [0, R_n^2]$, first transport membership to $[0, L_n]$, apply the hypothesis, and then transport the resulting inequality—including its denominators and centre—back to the expected form. The second transport is logically distinct from the first because the local predicate depends on the scale expression.

The recorded successful proof performs both operations. The last declarations-only candidate performs the membership transport and then returns the unchanged conclusion of `hratio`, whose displayed type still contains L_n . Its retained compiler message identifies the resulting mismatch. Both invoke the two relevant lemmas. The evidence therefore concerns application and representation alignment, not simple absence of premise retrieval.

4.3 Scalar closure and supplied consumer

With the transported hypothesis, the provided integral theorem gives

$$d_{ij} := |\overline{H}(P)_{ij} - H_0(\xi)_{ij}| \leq (e + d c) / R_n.$$

Since $d \geq 0$, $c \leq |c|$ gives $e + d c \leq e + d |c|$. Division by $R_n > 0$ preserves the inequality. With $Q = e + d |c|$ and the supplied `hsmall`,

$$d_{ij} \leq (e + d c) / R_n \leq Q / R_n \leq \rho.$$

No assertion that $c > 0$ or $Q > 0$ is needed. The uniform constants and threshold are chosen before the chart parameter, pulse, scale index and multiplier in the enclosing theorem; a finite set of numerical values cannot replace that quantifier order.

The fixed line

```
obtain {hd, hw, he} :=
  hmargin p hp (normalizedPair P) hclose
```

then consumes the generated proof. The remaining fixed code invokes `column_scale_bounds` and completes `ZeroOrderBounds`. Thus the accepted object serves as an actual premise of the next operation, but the downstream organisation was already supplied.

For clarity, the supplied algebra has the form $A = B \text{diag}(c_0, c_1)$ with $s_n > 0$, $0 < \ell \leq s_n c_j \leq v$, $|\det B| \geq \delta$, $|B_{ij}| \leq M$ and $(B^{-1}T_0)_j \geq \delta$. For $\zeta \geq 0$,

$$|\det(s_n A)| \geq \ell^2 \delta, |(s_n A)_{ij}| \leq v M,$$

$$[A^{-1}(\zeta T_0)]_j \geq (\delta/v) s_n \zeta.$$

Positive column masses follow from the assumptions. The last lower bound is zero when $\zeta = 0$. These are the kinds of zero-order margins required by the supplied result, not independent new bounds discovered in the experiment. All broader pulse, pressure, derivative, energy and global-singularity obligations remain outside its endpoint.

4.4 What links to the early witness, and what does not

The residual example shows failure of a selected state summary to determine all corrective actions. The `hclose` example shows that a logically available result still has to be instantiated and transported to the type a consumer requires. Both distinguish an available result from the operative conditions of its next use. They act on different mathematical objects and were not jointly randomised. The accepted `hclose` does not demonstrate the minimax lower bound experimentally; the bound does not prove the cause of the guided completion. Their relation is a carefully specified theoretical correspondence, not an already closed empirical mediation chain.

Supplementary Note 5. The actual comparison

The primary record is the retained handoff MAOFIELD_NS_HCLOSE_PRO_HANDOFF_20260920.zip. The prior draft's analysis reported 859 payload members and checked raw response, candidate and compiler identities. The current analysis rereads and checks the retained streams, templates and compiler records in a fresh output directory. It does not generate new answers or rerun Lean. Recomputed tables are compared with the preserved tables. The unmodified handoff and prior analyses remain in the private provenance collection. Source-mapped review copies are supplied directly, with transformations explicitly identified.

Supplementary Table 3 | The paired request sequence.

Condition	Request 1	Request 2	Request 3
Applicability-guided (raw label HISTORY)	Rejected candidate; 6,230 output tokens	Rejected candidate; 6,225	Accepted candidate; 8,326
Declarations-only (raw label STANDARD)	Length limit; 32,768; no visible body, no compilation	Rejected candidate; 3,181	Rejected candidate; 5,060

Input totals are 25,757 and 12,124, and output totals are 20,781 and 41,009, respectively. The comparison has a single selected target and seed block. Neither three attempts nor five compiles is a set of independent participants. All compiler failures, the initial truncation and the reported successful type/axiom record remain visible. Source-table plots describe these observations without success-rate percentages or significance markings.

The condition-specific instruction is an investigator-written strategy. It adds the prescription to use the scale equality across the needed representations, then the integral bound and a scalar comparison. It was not produced by a blind historical summariser; the final model did not perform history retrieval. Both conditions had the same actual lemma declarations and outer theorem, but not the same information or subsequent feedback. Ordinary prompt exposition, context use, stochasticity and unequal realised effort are not isolated from the observed contrast.

5.1 Evidence files

The following identifiers index unchanged original excerpts in the authors' internal EVIDENCE/ORIGINAL_EXCERPTS/ collection. Mathematical inputs, the six requests, five candidate bodies and compiler records required for reproduction are supplied separately in the review-support archive, with their provenance and any minimal transformations documented in PROVENANCE_MAP.json. Private investigator communications and unrelated machine records are not automatically included in that archive.

Supplementary Table 4 | Source locations and their roles.

Evidence IDs	Role
E01–E05	Scope, PI instruction, result boundary, parameters and exploratory design evolution
E06–E08	Exact shared outer template, common declarations and extra applicability instruction
E09–E11	Accepted generated body, result record and compiler output
E12–E14	Final comparison body and error output; first truncated response receipt
E15	Recorded plan, seeds and arm order

Evidence IDs	Role
E16	Original mathematical source file
E17	Selected actual PI instructions
E18–E19	Original client and direct-run implementation, retained for reading, not run in this rewrite
E20	Earlier PI statement of the mother question and theory-led route

The candidate bodies reproduced below are transcribed from the unchanged earlier Supplement. Display wrapping is typographical; the original bytes in the evidence/archive remain the executable source of record. The absence of a first comparison candidate is recorded, not filled using reasoning text.

5.2 Shared outer theorem

```

import Checkpoint.NodeGoal

noncomputable section
open Set
open NavierStokes NavierStokes.PrimaryCovarianceBounds

namespace Branch.Direct

theorem completed {X : Type*} [TopologicalSpace X] {K : Set X}
  (hK : IsCompact K) (H0 : X → Mat2) (T0 : X → Vec2)
  (hH0 : ∀ i j, ContinuousOn (fun p => H0 p i j) K)
  (hT0 : ∀ i, ContinuousOn (fun p => T0 p i) K)
  (hcone : ∀ p ∈ K, SmoothCovariance.StrictCone (H0 p) (T0 p))
  (vr vt : TorusInverse.Plane) (hdet : vr.1 * vt.2 - vr.2 * vt.1 ≠ 0)
  {r0 h a A b B E D : ℝ} (hr0 : 0 < r0) (hh : 0 ≤ h)
  (ha : 0 < a) (hA : 0 < A) (hb : 0 < b) (hE : 0 ≤ E) (hD : 0 ≤ D) :
  ∃ N : ℕ, 4 ≤ N ∧ ∃ detGap entryBound inverseLower : ℝ,
    0 < detGap ∧ 1 ≤ entryBound ∧ 0 < inverseLower ∧
    ∀ n : ℕ, N ≤ n → ∀ p ∈ K, ∀ P : Fin 2 → PartitionedCovariance.Pulse,
      (∀ j, PulseCovariance.PulseBounds (slotRadius r0 h n) a A b B (P j).ψ (P j).x) →
      (∀ j i v, v ∈ Icc 0 (ChartScales.slotLength r0 h n) →
        |(P j).t v i / (P j).x v - H0 p i j| ≤ E / ChartScales.slotLength r0 h n +
        D * |v - ChartScales.slotLength r0 h n / 2| / ChartScales.slotLength r0 h n) →
      ∀ zeta : ℝ, 0 ≤ zeta →
        ZeroOrderBounds (Real.sqrt (ChartScales.S n)) detGap entryBound inverseLower zeta
        (PartitionedCovariance.pairMatrix vr vt r0 (fun _ => ChartScales.timeCoefficient h n) P)
        (FlatCovariance.scaledTarget zeta (T0 p)) := by
  obtain (rho, delta, M, hrho, hdelta, hM, hmargin) := compact_model_margins hK H0 T0 hH0 hT0 hcone
  let kappa := PartitionedCovariance.nativePrefactor vr vt r0
  have hkappa : 0 < kappa := PartitionedCovariance.nativePrefactor_pos hdet hr0
  let lo := scalarLower kappa r0 a B
  let hi := scalarUpper kappa r0 A b
  have hlo : 0 < lo := scalarLower_pos hkappa hr0 ha
  have hhi : 0 < hi := scalarUpper_pos hkappa hr0 hA hb
  let Q := E + D * |PulseCovariance.concentrationConstant a A b B|
  have hQ : 0 ≤ Q := by dsimp [Q]; positivity
  obtain (N, hN4, hN) := eventually_slotRadius_large hr0 hh (max 1 (Q / rho))
  refine (N, hN4, lo ^ 2 * delta, max 1 (hi * M), delta / hi,
    by positivity, le_max_left __, div_pos hdelta hhi, ?_)
  intro n hn p hp P hP hratio zeta hzeta
  have hN4 : 4 ≤ n := hN4.trans hn
  have hr : 0 < slotRadius r0 h n := slotRadius_pos hr0 h n
  have hrlarge : Q / rho ≤ slotRadius r0 h n := (le_max_right __).trans (hN n hn)
  have hsmall : Q / slotRadius r0 h n ≤ rho := by
    apply (div_le_iff hr).mpr
    have ht := (div_le_iff hrho).mp hrlarge
    nlinarith
  have hclose (i j : Fin 2) : |normalizedPair P i j - H0 p i j| ≤ rho := by
    -- MODEL_PROOF_HERE
  obtain (hd, hw, he) := hmargin p hp (normalizedPair P) hclose
  have hscale (j : Fin 2) : lo ≤ Real.sqrt (ChartScales.S n) *
    pairScales kappa (fun _ => ChartScales.timeCoefficient h n) P j ∧
    Real.sqrt (ChartScales.S n) * pairScales kappa (fun _ => ChartScales.timeCoefficient h n) P j ≤
  hi :=
    chart_column_mass_bounds hr0 hh hkappa hN4 (hP j)
  have hR : 0 < Real.sqrt (ChartScales.S n) := Real.sqrt_pos.mpr (ChartScales.S_pos (by omega))
  have hbounds := column_scale_bounds (normalizedPair P) (T0 p)
    (pairScales kappa (fun _ => ChartScales.timeCoefficient h n) P)
  hR hlo hhi hdelta hzeta hscale hd hw he

```

```

rw [pairMatrix_factorization P hP vr vt r0]
exact (hbounds.1, fun i j => (hbounds.2.1 i j).trans (le_max_right _ _), hbounds.2.2)

end Branch.Direct

#check (Branch.Direct.completed : NavierStokes.PrimaryCovarianceBounds.checkpoint_bridge_goal)
#print axioms Branch.Direct.completed

```

5.3 Common lemma declarations

Both arms receive these identical existing declarations from `NavierStokes.PrimaryCovarianceBounds`. Their actual proofs are already available in the retained imported source. The fixed surrounding proof supplies the other parameters and assumptions.

```

theorem slotRadius_sq {r0 : ℝ} (hr0 : 0 < r0) (h : ℝ) (n : ℕ) :
  slotRadius r0 h n ^ 2 = ChartScales.slotLength r0 h n
noncomputable def normalizedPair (P : Fin 2 → PartitionedCovariance.Pulse) : Mat2 :=
  fun i j => PulseCovariance.normalizedColumn (P j).ψ (P j).x (P j).t i
theorem normalizedPair_entry_error {r a A b B E D : ℝ}
  (P : Fin 2 → PartitionedCovariance.Pulse)
  (hP : ∀ j, PulseCovariance.PulseBounds r a A b B (P j).ψ (P j).x)
  (H0 : Mat2) (hE : 0 ≤ E) (hD : 0 ≤ D)
  (hratio : ∀ j i v, v ∈ Icc 0 (r ^ 2) →
    |(P j).t v i / (P j).x v - H0 i j| ≤
      E / r ^ 2 + D * |v - r ^ 2 / 2| / r ^ 2) (i j : Fin 2) :
  |normalizedPair P i j - H0 i j| ≤
    (E + D * PulseCovariance.concentrationConstant a A b B) / r

```

5.4 Exact added guidance

The source result is `PrimaryCovarianceBounds.normalizedPair_entry_error`. It transfers the actual pulses' pointwise ratio-error premise to an error bound for their normalized integral matrix. Its exact declaration is already present in the common mathematical inputs.

To apply it at the current gap, match its pulse family to `P`, its model matrix to `H0 p`, and its nonnegative error coefficients to the existing `E`, `D`, `hE`, `hD`. The supplied `hP` has exactly the required `PulseBounds` type.

There is one representation mismatch in the pointwise premise: that result uses $(\text{slotRadius } r0 \ h \ n)^2$, whereas the existing `hratio` uses `ChartScales.slotLength r0 h n`. The established equality `slotRadius_sq hr0` rewrites these to the same interval, denominator and center; no new analytical estimate is needed for this step.

The result then bounds the local error by

$$(E + D * \text{concentrationConstant } a \ A \ b \ B) / \text{slotRadius } r0 \ h \ n.$$

The surrounding proof has already defined $Q = E + D * |\text{concentrationConstant } a \ A \ b \ B|$ and established `hsmall : Q / slotRadius r0 h n ≤ rho`. Use `concentrationConstant ≤ |concentrationConstant|`, $D \geq 0$ and the positive radius `hr` to pass from the result's numerator to Q . Transitivity then reaches the required ρ bound.

Remaining work: express this short application, rewrite and scalar comparison in Lean at the existing local goal. This note supplies no tactic script or completed proof body.

5.5 Guided: request 1

```

have hsl_eq : slotRadius r0 h n ^ 2 = ChartScales.slotLength r0 h n := slotRadius_sq hr0 h n
have hnorm := normalizedPair_entry_error P hP (H0 p) hE hD (by
  intro j i v hv
  have hv' : v ∈ Icc 0 (ChartScales.slotLength r0 h n) := by
    rw [hsl_eq]
  exact hv
  simp [hsl_eq] using hratio j i v hv')

```

```

) i j
calc
|normalizedPair P i j - H0 p i j| ≤ (E + D * PulseCovariance.concentrationConstant a A b B) /
slotRadius r0 h n := hnorm
_ ≤ (E + D * |PulseCovariance.concentrationConstant a A b B|) / slotRadius r0 h n := by
  apply (div_le_iff hr).mpr
  have := mul_le_mul_of_nonneg_left (le_abs_self (PulseCovariance.concentrationConstant a A b B))
hD
  linarith
_ = Q / slotRadius r0 h n := by dsimp [Q]
_ ≤ rho := hsmall

```

5.6 Guided: request 2

```

have hsl_eq : slotRadius r0 h n ^ 2 = ChartScales.slotLength r0 h n := slotRadius_sq hr0 h n
have hhratio_sq : ∀ j i v, v ∈ Icc 0 (slotRadius r0 h n ^ 2) → |(P j).t v i / (P j).x v - (H0 p) i j|
≤ E / slotRadius r0 h n ^ 2 + D * |v - slotRadius r0 h n ^ 2 / 2| / slotRadius r0 h n ^ 2 := by
  intro j i v hv
  have hv' : v ∈ Icc 0 (ChartScales.slotLength r0 h n) := by rw [hsl_eq]; exact hv
  have h := hratio j i v hv'
  rw [← hsl_eq] at h
  simp using h
have hnorm := normalizedPair_entry_error P hP (H0 p) hE hD hhratio_sq i j
calc
|normalizedPair P i j - H0 p i j| ≤ (E + D * PulseCovariance.concentrationConstant a A b B) /
slotRadius r0 h n := hnorm
_ ≤ (E + D * |PulseCovariance.concentrationConstant a A b B|) / slotRadius r0 h n := by
  apply (div_le_iff hr).mpr
  have := mul_le_mul_of_nonneg_left (le_abs_self (PulseCovariance.concentrationConstant a A b B))
hD
  linarith
_ = Q / slotRadius r0 h n := by dsimp [Q]
_ ≤ rho := hsmall

```

5.7 Guided: request 3

```

have hsl_eq : slotRadius r0 h n ^ 2 = ChartScales.slotLength r0 h n := slotRadius_sq hr0 h n
have hhratio_sq : ∀ j i v, v ∈ Icc 0 (slotRadius r0 h n ^ 2) → |(P j).t v i / (P j).x v - (H0 p) i j|
≤ E / slotRadius r0 h n ^ 2 + D * |v - slotRadius r0 h n ^ 2 / 2| / slotRadius r0 h n ^ 2 := by
  intro j i v hv
  have hv' : v ∈ Icc 0 (ChartScales.slotLength r0 h n) := by simp [hsl_eq] using hv
  have h := hratio j i v hv'
  simp [hsl_eq] using h
have hnorm := normalizedPair_entry_error P hP (H0 p) hE hD hhratio_sq i j
calc
|normalizedPair P i j - H0 p i j| ≤ (E + D * PulseCovariance.concentrationConstant a A b B) /
slotRadius r0 h n := hnorm
_ ≤ (E + D * |PulseCovariance.concentrationConstant a A b B|) / slotRadius r0 h n := by
  have hnum : E + D * PulseCovariance.concentrationConstant a A b B ≤ E + D * |
PulseCovariance.concentrationConstant a A b B| := by
    nlinarith [le_abs_self (PulseCovariance.concentrationConstant a A b B), hD]
  rw [div_le_iff hr]
  have hfield : ((E + D * |PulseCovariance.concentrationConstant a A b B|) / slotRadius r0 h n) *
slotRadius r0 h n = E + D * |PulseCovariance.concentrationConstant a A b B| := by
    field_simp [hr.ne']
  rw [hfield]
  exact hnum
_ = Q / slotRadius r0 h n := by dsimp [Q]
_ ≤ rho := hsmall

```

5.8 Declarations only: request 1

No visible proof body. Recorded finish reason: length. No candidate was compiled.

5.9 Declarations only: request 2

```

have hsq : slotRadius r0 h n ^ 2 = ChartScales.slotLength r0 h n := slotRadius_sq hr0 h n
have hentry : |normalizedPair P i j - H0 p i j| ≤
  (E + D * PulseCovariance.concentrationConstant a A b B) / slotRadius r0 h n :=
  normalizedPair_entry_error (slotRadius r0 h n) a A b B E D P hP (H0 p) hE hD
  (fun j' i' v' hv' => hratio j' i' v' (by simp [hsq] using hv')) i j

```

```

have habs : D * PulseCovariance.concentrationConstant a A b B ≤ D * |
PulseCovariance.concentrationConstant a A b B| := by
  nlinarith [le_abs_self (PulseCovariance.concentrationConstant a A b B), hD]
have hle1 : (E + D * PulseCovariance.concentrationConstant a A b B) / slotRadius r0 h n ≤
  Q / slotRadius r0 h n := by
  dsimp only [Q]
  rw [div_le_div_iff (le_of_lt hr) (le_of_lt hr)]
  nlinarith [habs]
exact hentry.trans (hle1.trans hsmall)

```

5.10 Declarations only: request 3

```

have hsq : slotRadius r0 h n ^ 2 = ChartScales.slotLength r0 h n := slotRadius_sq hr0 h n
have hentry : |normalizedPair P i j - H0 p i j| ≤
  (E + D * PulseCovariance.concentrationConstant a A b B) / slotRadius r0 h n :=
  normalizedPair_entry_error (r := slotRadius r0 h n) P hP (H0 p) hE hD
  (fun j' i' v' hv' => by
    have hv'' : v' ∈ Icc 0 (ChartScales.slotLength r0 h n) := by
      simpa [hsq] using hv'
    exact hratio j' i' v' hv'') i j
have hle1 : (E + D * PulseCovariance.concentrationConstant a A b B) / slotRadius r0 h n ≤
  Q / slotRadius r0 h n := by
  dsimp only [Q]
  gcongr
  nlinarith [le_abs_self (PulseCovariance.concentrationConstant a A b B), hD]
exact hentry.trans (hle1.trans hsmall)

```

Supplementary Note 6. Historical results stay historical

The following material is not pooled with `hclose`. Some rows are historical reports, rather than raw-data re-audits available in the present writing round. `NOT_RUN` in a proposal refers to that proposal's date. A later record is separately attributed; it does not alter the earlier document's identity. Historical protocols remain distinct from subsequent analysis.

Supplementary Table 5 | Retained episodes and interpretation.

Source or episode	Retained observation or status	What this paper may infer
Early paused software A	Existing repair retrieved from Git; first completed pair equal. Second unit incomplete, with PI pause and unknown/unrun branches	A documented sufficient-route/zero and incomplete-case boundary, not universal failure or success
A' V1, historical review	Four lineages, 48 stages; deployed programme unchanged; primary deployment contrasts zero. Interface defects separately found	The product readout and process changes were distinct; defects were not proved to cause all zeros
A'' stopped, historical report	Nine submitted stages, one no-submission stop, one interruption, 21 unrun; long repetition and task/ACK issues	Termination and unrun states stay distinct from mathematical error; no claim of an identified intrinsic loop cause
E1/E3 repaired R2	Eight completed stages and 56 readable coordinates; four E3 accounts contained record/count discrepancies; semantic disagreements remained	Correct labels or complete JSON do not certify accurate evidence binding; not a broad causal effect
Early NS coefficient R4 A	Complete bounded run; post-adaptation paired differences zero or rounding scale; cold execution mixed	Legal rederivation/convergence remains possible; cold advantages do not replace the main endpoint
Coefficient-study proposals	Conditional mathematics and planned studies, unrun at their issue dates	Design documents only; their status is not substituted for the later completed runs

Source or episode	Retained observation or status	What this paper may infer
Completed coefficient assistance/adoption studies	20 + 32 recorded stages; 104 complete returns; 40 checked inherited-state arrays; actual revision/reuse; five empty finals and seven identical adoption pairs	Process observations and original mixed contrasts; no established stable average benefit. Full details in Note 9
Broader full-proof/history proposals	Source builds, original discovery snapshots, kernel checks or intervention branches remained NOT_RUN/UNBOUND where reported	Not a completed reconstruction of OpenAI's discovery; not evidence of history-compression effects

The final handoff's pre-hclose evolution provides more specific workflow records:

Supplementary Table 6 | Earlier workflow identities.

Version or activity	Reported scope	Reported result	Current status
Full-endpoint workflow	Four full-endpoint branches	55 requests; zero writes	Earlier exploration
Adapted continuation	First adapted branch	40 requests; zero writes; PI stopped	Stopped exploration
Dynamic organisation	First dynamically organised branch	23 requests; zero writes	Earlier exploration
Material preparation	Material preparation	16 calls; packet not accepted; four branches unrun	Preparation failure
Fixed-material continuation	Fixed-material continuation	23 requests; zero writes; remaining three unrun	Earlier exploration
Precompiled-source continuation	Precompiled common source	First history branch: 42 requests, zero writes; stop	Stopped exploration
Independent Codex diagnostic	Own specified same-start task and materials; different model/tool environment	Saved local compilation records; interrupted	Not a formal effect control; blindness not independently certified
Reference-exposed researcher recheck	Complete reference already seen	Algebra derivation and checks; explicitly no new Lean compilation	Researcher verification, not independent discovery
Final hclose pair	One selected hole, common enclosing proof	Guided third candidate accepted; comparison incomplete within three	Present local evidence

“Zero writes” is a description of a workflow record, not a general statement of model incapacity. Earlier resource changes, PI stopping and outcome-informed selection are part of the history. The final narrow success does not retrospectively complete any broader endpoint. The accompanying evidence ledger records source paths for historical statements; unavailable raw streams remain unavailable rather than being inferred from summaries.

Supplementary Note 7. Attribution and further research

7.1 U1, U2 and U3

The source lemma is an existing method: its use is U1. The successful experiment produced a new token-level proof object in response to researcher guidance and compiler feedback. This is an observed local construction, not evidence that the model originated a new general method. Same-goal candidate correction is not by itself U3. The supplied `hmargin` consumer documents use but does not alone

establish changed-duty reconstruction. The separate coefficient records in Note 9 do include changed targets, actual reconstruction and renewed use within a supplied interface. This observed restricted process must not be confused with an unestablished general causal benefit or autonomous formation of the whole research arrangement.

The human investigator selected and changed the task, supplied the application strategy and authorised execution. The agent constructed tactic bodies and the compiler produced feedback. This division is recorded and attributable. We do not infer that the investigator’s understanding improved, that a particular subjective state caused success, or that human involvement is inherently necessary in all systems. A single integrated solver or a sufficient compact interface may realise the same future function.

7.2 Observed continuation and the remaining attribution question

Changed-target continuation was executed in the coefficient studies; it is not a missing experiment in the sense of never having been run. The records show actual state inheritance, particular programme revisions and subsequent application. The original comparisons nevertheless failed to establish stable average benefits, for reasons detailed in Note 9. Nor did the separate proof-completion pair test equal accepted present proof artefacts under a new duty. These are different limitations and should not be merged.

A future bounded comparison could hold an actually accepted proof artefact and ordinary source access fixed, predefine a new representation or assumption-handling duty, and compare an application-bearing handoff with normal compact continuation. The reconstructed artefact would then have to be used in one further action. Such a study would address the remaining same-object attribution question, not supply the first observation that any successor use exists. It would retain legitimate rederivation, equality, failure and unknown completion without an ability screen. No additional model sampling is performed for the present revision.

7.3 Formal and publication responsibilities

The current acceptance rests on retained compiler evidence and the earlier source audit. Fresh kernel checking, an external mathematical judgement of the intended target and public release permissions remain separate tasks, not results of this writing. They are not additional scientific effect experiments proposed here. The original NS work remains attributed to its source authors. Publication priority for the present conceptual synthesis or local analytical use has not been independently established.

Supplementary Note 8. Reproducibility layers

Original evidence is preserved unchanged. Deterministic reanalysis and saved-candidate preparation are recorded in separate output directories and do not add experimental units. The review-support archive provides direct access to mathematical inputs, candidate mappings, Source Data and reproduction commands; its privacy and provenance map identifies the minimal transformations applied to shareable records.

Supplementary Table 7 | Reproducibility layers and observed status.

Evidence or check	Actual identity in this analysis
Analytical separation and sufficient repair	Written proofs in Notes 2–3; standard algebra and state composition
Exact algebra implementation	REPRODUCIBILITY/math/exact_checks.py, executed locally; output and logs retained

Evidence or check	Actual identity in this analysis
Six-request record analysis	Private full-record audit and source-mapped visible-record audit both reread streams, templates and compiler receipts; no new sampling
Coefficient-continuation reanalysis	Complete terminal rows and parent-state arrays reread; selected actual expressions evaluated separately; no new model sampling
Independent diagnostic	32_INDEPENDENT_DIAGNOSTIC; its own task boundary and retained local compilation
Researcher recheck	33_RESEARCHER_RECHECK; reference exposed, algebra only, no new Lean compilation in that record
Saved five-candidate replay	Source assembly executed for five unmodified candidates; Lean/Lake/Docker unavailable here, compilation replay NOT_RUN
New model sampling / external kernel / external review	NOT_RUN

In the internal evidence package, `python REPRODUCIBILITY/reproduce.py --out WORK/reproduction` verifies all 859 original payloads, reconstructs six returned requests and five candidates, calculates totals and runs exact algebra checks. In the review-support archive, the same entry point recomputes the corresponding tables from the shareable, source-mapped request, visible-response and compiler records. Neither entry contacts a model endpoint. `python REPRODUCIBILITY/experiment/assemble_lean.py --out WORK/lean_project` assembles the preserved workspace with its public modules, including `Checkpoint/NodeGoal.lean`, and verifies the saved candidate/request mapping. `python REPRODUCIBILITY/experiment/replay_lean.py --probe --prepared-project WORK/lean_project --out WORK/lean_probe` records toolchain availability. Compilation requires an explicit execute option and the isolated, dependency-ready environment described in the assembly README; it is not inferred from script availability.

The mathematical witness and `hclose` are different subproblems. The former proves a finite observation restriction and its compact repair. The latter records instantiation and transport of an existing theorem, construction of a specific proof object, and use by a supplied consumer. The coefficient records additionally observe changed-target reconstruction and renewed use. None of these evidence layers alone identifies a general mechanism governing all jointly formed research conditions.

Supplementary Note 9. Executed successor evidence

9.1 Source, population and analysis identity

The coefficient studies are separate from the proof-completion pair. The unchanged archive is `NS_R6_AB_RAW_WITH_A_SUPPLEMENT_20260919_UNDER100MB.tar.xz`, SHA-256 `9b8fde263413edceb5a980de182a347e4aa10995afc20c982e8e8f78dcb43ba4`. The review-support files retain source-member paths and original hashes. No new model response is included in this reanalysis.

The assistance study has four formation episodes descended from two archived ancestors, 20 recorded stages and 40 complete returns. The version-adoption study has eight new formation episodes, 32 stages and 64 complete returns. All 104 complete returns have a recorded stop; a previous assistance request remains unknown. Its separately authorised supplementary observation is not the recovery of the original answer. The assistance first-continuation main endpoint was unaffected by that supplementary second-continuation observation. These version identities are retained, not merged into a homogeneous original run.

Each programme supplies two expressions $B(r, u)$ and $C(r, u)$ to a common operation

$$g = r/2 + \Phi(r, u)[r^2 B(r, u) + ur C(r, u)], E = (z + g)^2 - p.$$

The cutoff equals one in the stated safe core and zero outside the stated outer rectangle. Targets change according to the original fixed rules; the next source is calculated from the actual preceding z , not an ideal solution. The operator template, source families, feedback and subsequent duties are supplied by the investigators. Numerical validity is not a certificate of complete PDE admissibility.

The present audit rereads all terminal programme and field files, checks the arithmetic of 18,720 terminal rows, matches 40 inherited arrays to their actual predecessor output strings and hashes, and evaluates nine displayed programme/coordinate combinations using a separate restricted expression interpreter at 100-digit precision. Stored 70-digit rounding is retained as recorded, not replaced by exact zero. The audit entry point is REPRODUCIBILITY/successor/audit_successor.py; its raw export and source-mapped reanalysis are separated.

9.2 All original comparisons and null information

The following differences are the original family-wise mean absolute residual in treatment minus control, after adaptation. The assistance treatment is structural feedback rather than ordinary numerical feedback; adoption compares the actual revised and initial programmes at the same first-continuation state. All planned formation episodes are in the denominators. Rows are repeated measurements, not independent histories.

Supplementary Table 8 | Original complete-denominator contrasts.

Source family	Assistance: first continuation	Assistance: second continuation, supplementary view	Adoption: second continuation
ZERO	0	0	0
UNIFORM	4.362002266e-06	-7.74210004e-11	-3.43616067e-08
LEFT	-224.6653146	-1020169.24	1020169.24
RIGHT	-224.6653322	-1020169.655	1020169.655
BOTH	-282.4465954	-1497107.113	1497107.113

Assistance includes 144 paired positions per source family from four formation episodes and two archived ancestors; adoption includes 288 from eight new episodes. A negative difference denotes a smaller value of this original unweighted observation, not universally better scientific performance. Zero-source outcomes remain zero. In the adoption study, initial and revised expressions were identical in seven of eight episodes. Those episodes remain included. In the single different-version episode, the initial-version branch could and did recover the accessible revised expression in its draft, before its final visible response was empty.

There is one empty final programme in assistance and four in adoption. All four adoption cases already had executable drafts; the protocol used only the final response to decide whether to apply the new update. The final-response semantics are preserved rather than retrospectively replacing empty responses with drafts. Outside the cutoff, all valid programmes apply the same linear increment while absent programmes apply none:

$$E_{linear} = ur + r^2/4, E_{no\ update} = -r,$$

$$E_{linear} - E_{no\ update} = z r + r^2 / 4.$$

Consequently, extreme outer positions can dominate differences without depending on the submitted B, C . At one original position, repeated outer updates take the amplitude approximately through $1, -15, -143, -10383$. That is an execution property of this finite workbench, not a physical NS singularity. It explains why the large family means are not evidence of large mathematical assistance benefits or of harmful revised knowledge. No empty case is removed to compute a preferred causal estimate, and no alternative weighting replaces the original primary observation. The record therefore establishes neither a stable average assistance benefit nor a stable adoption advantage.

9.3 First actual revision: preserving a method, changing its use

The following trace was selected after inspection as an explanatory example, not a confirmatory unit. Source identifiers are assistance P00, first continuation slot 0 and its actual successor. The existing programme uses

$$B_0 = -\frac{1}{2(1+\sqrt{1+r})^2}, C_0 = 0.$$

Let $s = \sqrt{1+r}$. In the safe core, $g_0 = s - 1$ has zero pure residual, while $E_0 = 2u(s - 1)$. The first generated revision is

$$C_1 = -\frac{1}{s(1+s)}.$$

Then $z + g_1 = s + u/s$, and direct subtraction of the target $s^2 + 2u + u^2$ gives

$$E_1 = -\frac{u^2 r}{1+r}.$$

Following actual feedback, the final programme retains B_0 but sets

$$w = \sqrt{(1+u)^2 + r}, C_2 = -\frac{1/(w+1+u) + 1/(s+1)}{w+s}.$$

Rationalising $w^2 - (1+u)^2 = r$ and $s^2 - 1 = r$ yields

$$ur C_2 = w - s - u,$$

so $g_2 = w - (1+u)$ and $E_2 = 0$ in the core. The formula does not divide by ur and extends across $u = 0$ or $r = 0$. On the outer closed rectangle, $s \geq 1/\sqrt{2}$, $1+u \geq 3/4$ and $w \geq 1/4$, so its denominators stay positive. This is a local smooth formula, not the non-smooth clipped root across a negative radicand.

At one original feedback position (uniform family, $\tau = 1/64$, $x = 1/8$, slot 0), the initial, draft and final whole residuals are approximately 1.5011272×10^{-6} , $-1.1680410 \times 10^{-8}$ and 1.8111×10^{-71} . The final value is stored arithmetic roundoff. The final pure and inherited-state terms are approximately $-1.4895364 \times 10^{-6}$ and $+1.4895364 \times 10^{-6}$. Thus restoring the whole is not the same as keeping the previously exact pure component at zero. The next continuation received the real final programme and the actual resulting state, and retained that programme for its changed target. This is actual reuse, not a proposed successor run.

9.4 Second actual revision: state terms across two duties

In adoption episode P07, the actual sequence preserves $B = -1/8$ and changes

$$C_0 = -1/2, C_1 = -1/2 + u/2, C_2 = -1/2 + u/2 - u^2/2.$$

With $g_j = r/2 - r^2/8 + urC_j$, expansion for $u = a\tau, r = b\tau^3$ gives

$$E_{C_0} = -a^2 b \tau^5 + O(\tau^7),$$

$$E_{C_1} = a^3 b \tau^6 + O(\tau^7),$$

$$E_{C_2} = -(a^4 b + 3ab^2/4)\tau^7 + O(\tau^8).$$

These are standard series relations evaluated for the generated programmes. They are not a claim of mathematical priority or uniform improvement: vanishing leading coefficients change the order, and for $r = b\tau^2$ the two later programmes generally retain an order- $\tau^5 ur^2$ channel.

On the same prestate within the first continuation, the recorded old/new residuals are approximately -1.19294×10^{-8} and -1.26328×10^{-10} . On the distinct prestate within the second continuation they are -5.37347×10^{-13} and -2.72164×10^{-15} . Figure 3 keeps these within-duty pairs separate. The second duty genuinely consumes the first one's stored field and programme. It is not a rerun from the initial amplitude. A sign error in the generated prose about the remaining ur^2 term is retained in the source; the expression's whole expansion has a negative coefficient. Valid execution and an accurate explanation are separate claims.

9.5 What the combined evidence establishes

The coefficient records show actual formation, feedback, state-dependent revision and renewed use under new fixed targets. The hclose pair separately shows a shared result being converted into an applicable proof object and used by a supplied consumer. Neither justifies a claim that the entire human-AI research cycle, a stable average historical benefit, autonomous problem formation or the OpenAI discovery process has been causally identified. These limitations do not erase executed continuation. They locate the positive process observations and the unresolved comparative claims at their correct levels.